

Docker Volumes & Data Persistence

Practical Assignment 8

Step 1:

```
Command Prompt
::1/128
Live Restore Enabled: false
Firewall Backend: iptables

C:\Users\samik>docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:7f4da0fc94bcece205a8c0b6f4d11c8196924654ffe5c4d1aa439b7f632048b2
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

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C:\Users\samik>docker run -dit --name data-test ubuntu
601e1f0ae4b2f459c87f4597fa70d1b4fcdela3d5798861515a0784c735e887
C:\Users\samik>
```

Step 2:

```
C:\Windows\System32\cmd.exe
D:\>docker --version
Docker version 29.4.0, build 9d7ad9f

D:\>docker info
Client:
 Version: 29.4.0
 Context: desktop-linux
 Debug Mode: false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version: v1.42.0
    Path: C:\Program Files\Docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version: v1.20.2
    Path: C:\Program Files\Docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version: v0.33.0-desktop.1
    Path: C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version: v5.1.1
    Path: C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version: 0.0.47
    Path: C:\Program Files\Docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Docker Inc.)
    Version: v0.3.0
    Path: C:\Program Files\Docker\cli-plugins\docker-desktop.exe
  dhi: CLI for managing Docker Hardened Images (Docker Inc.)
    Version: v0.0.2
    Path: C:\Program Files\Docker\cli-plugins\docker-dhi.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version: v0.2.31
    Path: C:\Program Files\Docker\cli-plugins\docker-extension.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version: v1.4.0
    Path: C:\Program Files\Docker\cli-plugins\docker-init.exe
  mcp: Docker MCP Plugin (Docker Inc.)
    Version: v0.40.3
    Path: C:\Program Files\Docker\cli-plugins\docker-mcp.exe
  model: Docker Model Runner (Docker Inc.)
```

Step3:

```
C:\Windows\System32\cmd.exe
Path: C:\Program Files\Docker\cli-plugins\docker-extension.exe
init: Creates Docker-related starter files for your project (Docker Inc.)
Version: v1.4.0
Path: C:\Program Files\Docker\cli-plugins\docker-init.exe
mcp: Docker MCP Plugin (Docker Inc.)
Version: v0.40.3
Path: C:\Program Files\Docker\cli-plugins\docker-mcp.exe
model: Docker Model Runner (Docker Inc.)
Version: v1.1.29
Path: C:\Program Files\Docker\cli-plugins\docker-model.exe
offload: Docker Offload (Docker Inc.)
Version: v0.5.82
Path: C:\Program Files\Docker\cli-plugins\docker-offload.exe
pass: Docker Pass Secrets Manager Plugin (beta) (Docker Inc.)
Version: v0.0.25
Path: C:\Program Files\Docker\cli-plugins\docker-pass.exe
sandbox: Docker Sandbox (Docker Inc.)
Version: v0.12.0
Path: C:\Program Files\Docker\cli-plugins\docker-sandbox.exe
sbom: View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc.)
Version: 0.6.0
Path: C:\Program Files\Docker\cli-plugins\docker-sbom.exe
scout: Docker Scout (Docker Inc.)
Version: v1.20.3
Path: C:\Program Files\Docker\cli-plugins\docker-scout.exe

Server:
Containers: 22
  Running: 6
  Paused: 0
  Stopped: 16
Images: 13
Server Version: 29.4.0
Storage Driver: overlayfs
  driver-type: io.containerd.snapshotter.v1
Logging Driver: json-file
Cgroup Driver: cgroupfs
Cgroup Version: 2
Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
```

Step 4:

```
C:\Windows\System32\cmd.exe
Volume: local
Network: bridge host ipvlan macvlan null overlay
Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
CDI spec directories:
  /etc/cdi
  /var/run/cdi
Discovered Devices:
  cdi: docker.com/gpu=webgpu
Swarm: inactive
Runtimes: io.containerd.runc.v2 nvidia runc
Default Runtime: runc
Init Binary: docker-init
containerd version: dea7da592f5d1d2b7755e3a161be07f43fad8f75
runc version: v1.3.4-0-gd6d73eb8
init version: de40ad0
Security Options:
  seccomp
  Profile: builtin
  cgroupns
Kernel Version: 6.6.87.2-microsoft-standard-WSL2
Operating System: Docker Desktop
OSType: linux
Architecture: x86_64
CPUs: 12
Total Memory: 7.573GiB
Name: docker-desktop
ID: c2a6df9c-9d80-4e5b-b40d-f839fa201baa
Docker Root Dir: /var/lib/docker
Debug Mode: false
HTTP Proxy: http.docker.internal:3128
HTTPS Proxy: http.docker.internal:3128
No Proxy: hubproxy.docker.internal
Labels:
  com.docker.desktop.address=npipe://\\.\pipe\docker_cli
Experimental: false
Insecure Registries:
  hubproxy.docker.internal:5555
  127.0.0.0/8
  ::1/128
Live Restore Enabled: false
Firewall Backend: iptables
```

Step 5:

```
C:\Windows\System32\cmd.exe x + v

D:\>docker run hello-world

Hello from Docker!
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D:\>docker run -dit --name data-test ubuntu

What's next:
  Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: Conflict. The container name "/data-test" is already in use by container "601e1f0ae4b2f459c87f4597fa70d1b4fcde1a3d579088
61515a0784c735e887". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information

D:\>docker exec -it data-test bash
root@601e1f0ae4b2:/# mkdir /mydata
root@601e1f0ae4b2:/# echo "This is my important data" > /mydata/student.txt
root@601e1f0ae4b2:/# cat /mydata/student.txt
This is my important data
root@601e1f0ae4b2:/# exit
exit
```

Step 7:

```
C:\Windows\System32\cmd.exe x + v

D:\>docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
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docker: Error response from daemon: Conflict. The container name "/data-test" is already in use by container "601e1f0ae4b2f459c87f4597fa70d1b4fcde1a3d579088
61515a0784c735e887". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information

D:\>docker exec -it data-test bash
root@601e1f0ae4b2:/# mkdir /mydata
root@601e1f0ae4b2:/# echo "This is my important data" > /mydata/student.txt
root@601e1f0ae4b2:/# cat /mydata/student.txt
This is my important data
root@601e1f0ae4b2:/# exit
exit
```

Step 8:

```
C:\Windows\System32\cmd.exe
root@601ef0ae4b2:/# exit
exit

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug data-test
Learn more at https://docs.docker.com/go/debug-cli/

D:\>mkdir /mydata
The syntax of the command is incorrect.

D:\>docker exec -it data-test bash
root@601ef0ae4b2:/# mkdir /mydata
mkdir: cannot create directory '/mydata': File exists
root@601ef0ae4b2:/# echo "This is my important data" > /mydata/student.txt
root@601ef0ae4b2:/# cat /mydata/student.txt
This is my important data
root@601ef0ae4b2:/# exit
exit

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug data-test
Learn more at https://docs.docker.com/go/debug-cli/

D:\>docker rm -f data-test
data-test

D:\>docker run -dit --name data-test2 ubuntu
88fdf27523326ed2a5b9ce6566408227fddb0f636b80ab671bd1e43003824dd

D:\>docker exec -it data-test2 bash
root@88fdf2752332:/# ls /mydata
ls: cannot access '/mydata': No such file or directory
root@88fdf2752332:/# exit
exit

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug data-test2
Learn more at https://docs.docker.com/go/debug-cli/

D:\>docker rm -f data-test2
data-test2
```

Step 9:

```
C:\Windows\System32\cmd.exe
D:\>docker rm -f data-test2
data-test2

D:\>docker volume create student-data
student-data

D:\>docker volume ls
DRIVER      VOLUME NAME
local       3a3e1b3985ba8c3a22aaaa4e647e780726a1e802abfa26a4e4b58b78dc7bc8
local       5da2b9c1e5287e3ffe0fc74da5ba0ffacfa584a22fb2a179a6077f43a9ceae60
local       8b530175afdc80bc0e836323af944589c9339ded8e18d7eb63238f53512a4bbe8
local       35bb8bd04eb3704ccf74e1d02142db2ed83ed2c4d9b921b5401fc08ba8a6759f
local       53dd083a8c9ae280417b5bd7a00476e58cb8f98fb14cf3f0668cf3ff6ace13fe
local       101a9b1bcae01f896cf9214231f5d5eed9903896a1e369574f00dc68641de88d
local       340ca122d046da8d6df0ebbdd729bf9c6b47db5386d27aedfb2860e5b0ca2a93
local       596eee4a51b6c0e08216cf7a3e37c8028ee973b898d227bb98f2927ff8695498
local       75118cfd6a0471b494d84a528f620ea17459ea79a81e0320288e576b54330171
local       156037c3408319a7d40cca85db22592fff3e2e00453905881e9e67a6cac62a74
local       7348025fbecedcd371cbde3731f9e4c76f6ddc65459354031e4381123a53ffb3
local       attendance-management_mongo_data
local       c3cb16d445a25a9f0997807d97db5bb901b308e6ef6f20bdf5ff1bf7976754083
local       def13087a632c59b43d2d81fea4d9ae5540435bc5fe6e2f3278fc1e076310642
local       jenkins_home
local       redis-data
local       student-data

D:\>docker volume inspect student-data
[
  {
    "CreatedAt": "2026-08-08T05:28:47Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/student-data/_data",
    "Name": "student-data",
    "Options": null,
    "Scope": "local"
  }
]
```

Step 10:

```
C:\Windows\System32\cmd.exe
D:\>docker run -dit --name volume-test -v student-data:/data ubuntu
0130778ab46f306d809c7530274a70fd198d0b7de528dd7f62683a485a1242a5

D:\>docker exec volume-test bash -c "echo 'Docker Volume Practical' > /data/notes.txt"

D:\>docker exec volume-test bash -c "echo 'Student Name: YourName' > /data/student.txt"

D:\>docker exec volume-test ls /data
notes.txt
student.txt

D:\>docker rm -f volume-test
volume-test

D:\>docker run -dit --name volume-test2 -v student-data:/data ubuntu
ea2df88624338fc4b6a744493ecce9d5195f023509bd5795fd6157e5317bb1dc

D:\>docker exec volume-test2 cat /data/notes.txt
Docker Volume Practical

D:\>docker exec volume-test2 cat /data/student.txt
Student Name: YourName

D:\>docker rm -f volume-test2
volume-test2

D:\>docker run -dit --name container1 -v student-data:/data ubuntu
484083ca9979d4f6913e1313305d0d80a48fe19745a9a53b91c213469fcbc7b2

D:\>docker run -dit --name container2 -v student-data:/data ubuntu
9e238383211922704973aa5310b709ec8c3dd047293445da587625db41794d40

D:\>docker exec container1 bash -c "echo 'Created by Container 1' > /data/container1.txt"

D:\>docker exec container2 cat /data/container1.txt
Created by Container 1

D:\>|
```

Step 11:

```
C:\Windows\System32\cmd.exe
9e238383211922704973aa5310b709ec8c3dd047293445da587625db41794d40

D:\>docker exec container1 bash -c "echo 'Created by Container 1' > /data/container1.txt"

D:\>docker exec container2 cat /data/container1.txt
Created by Container 1

D:\>docker run -dit --name mount-test --mount source=student-data,target=/data ubuntu
09a84c1e840432c2b7542d445585e7f702e7acd347318f59185527e02a734b43

D:\>docker exec mount-test ls /data
container1.txt
notes.txt
student.txt

D:\>--mount source=student-data,target=/data
'--mount' is not recognized as an internal or external command,
operable program or batch file.

D:\>docker run -dit --name readonly-test -v student-data:/data:ro ubuntu
742242787b19cfe6aec02e0823eed3b2f0ed70300e59be6747e464a9a3f3ceec

D:\>docker exec readonly-test bash -c "echo 'new data' > /data/newfile.txt"
bash: line 1: /data/newfile.txt: Read-only file system

D:\>docker rm -f mount-test readonly-test
mount-test
readonly-test

D:\>mkdir C:\DockerLabs\data

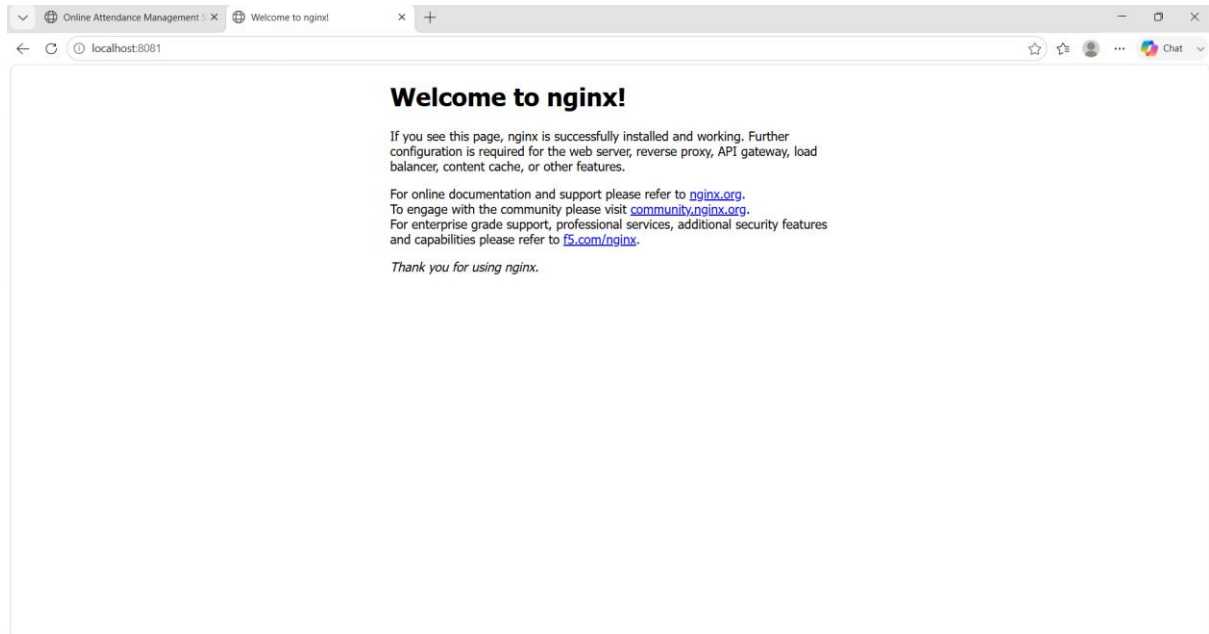
D:\>echo Hello from Windows > C:\DockerLabs\data\windows.txt

D:\>docker run -dit --name bind-test -v C:\DockerLabs\data:/data ubuntu
72817d30be1c4b443e6dc66b3372a907c126f21e82baa4c260b1d3b33bf34d36

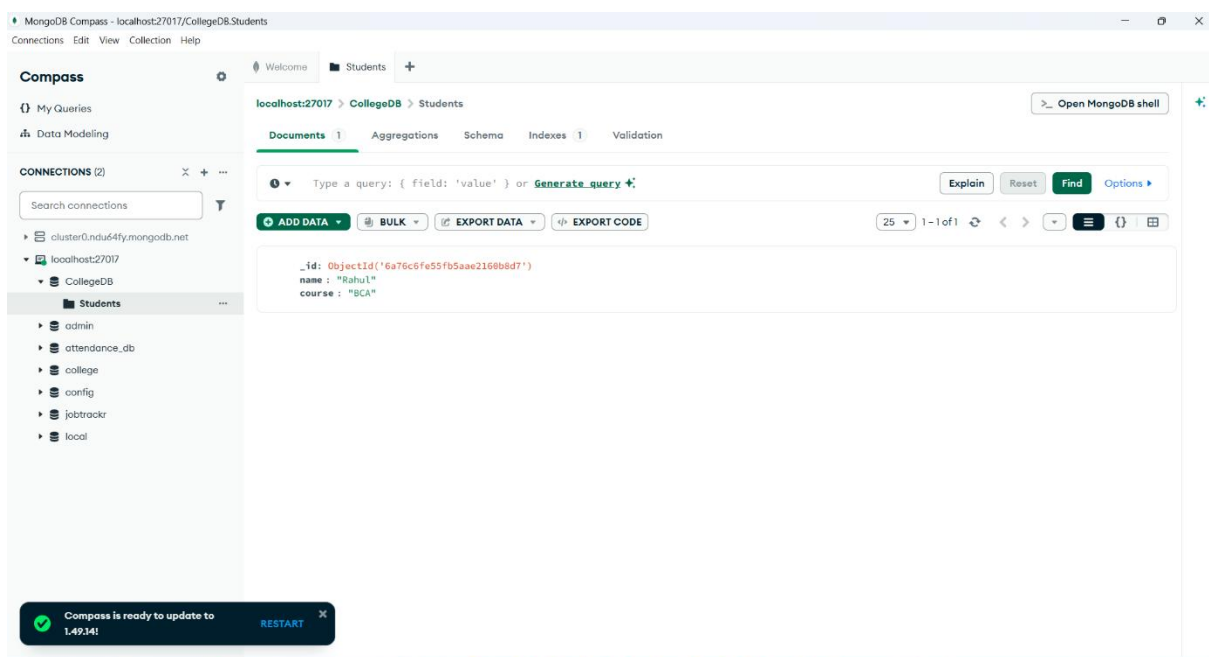
D:\>docker exec bind-test cat /data/windows.txt
Hello from Windows

D:\>docker exec bind-test bash -c "echo 'Modified from Docker' > /data/docker.txt"
```

Step 12:



Step 13:



Step 14:

```
C:\Windows\System32\cmd.exe
D:\>docker rm -f nginx-volume
nginx-volume

D:\>docker run -d --name nginx-volume -p 8081:80 -v website-data:/usr/share/nginx/html nginx
268bfd23dd5a5d87ef13677d6499288d5c4a8c926391336f9ab8a5ca7a429251

D:\>docker volume create mongo-data
mongo-data

D:\>docker run -d --name mongod -p 27017:27017 -v mongo-data:/data/db mongo

What's next:
  Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: Conflict. The container name "/mongod" is already in use by container "01be36fe5836f5a3ddf2a4272c17829703361a9e9f8b3089ba0783a444dd2890". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information

D:\>docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
268bfd23dd5a   nginx                               "/docker-entrypoint..." 6 minutes ago  Up 6 minutes  0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
72817d30be1c   ubuntu                             "/bin/bash"              16 minutes ago Up 16 minutes
9e2383832119   ubuntu                             "/bin/bash"              24 minutes ago Up 24 minutes
484083ca9979   ubuntu                             "/bin/bash"              24 minutes ago Up 24 minutes
83176449df47   attendance-management-frontend     "/docker-entrypoint..." 4 days ago    Up About an hour (unhealthy) 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
195bb2fc9bce   attendance-management-backend      "docker-entrypoint.s..." 4 days ago    Up About an hour (healthy) 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
5000/tcp
b1841cfcc7cb   attendance_backend                  "docker-entrypoint.s..." 4 days ago    Up About an hour (healthy) 0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
7->27017/tcp
mongo:7

D:\>
D:\>mkdir C:\DockerLabs\backup
```

Step 15:

```
C:\Windows\System32\cmd.exe
268bfd23dd5a   nginx                               "/docker-entrypoint..." 6 minutes ago  Up 6 minutes  0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
72817d30be1c   ubuntu                             "/bin/bash"              16 minutes ago Up 16 minutes
9e2383832119   ubuntu                             "/bin/bash"              24 minutes ago Up 24 minutes
484083ca9979   ubuntu                             "/bin/bash"              24 minutes ago Up 24 minutes
83176449df47   attendance-management-frontend     "/docker-entrypoint..." 4 days ago    Up About an hour (unhealthy) 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
195bb2fc9bce   attendance-management-backend      "docker-entrypoint.s..." 4 days ago    Up About an hour (healthy) 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
5000/tcp
b1841cfcc7cb   attendance_backend                  "docker-entrypoint.s..." 4 days ago    Up About an hour (healthy) 0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
7->27017/tcp
mongo:7

D:\>
D:\>mkdir C:\DockerLabs\backup

D:\>docker run --rm -v student-data:/data -v C:\DockerLabs\backup\:/backup ubuntu tar czf /backup/student-data.tar.gz -C /data .

D:\>docker volume create restored-data
restored-data

D:\>docker run --rm -v restored-data:/data -v C:\DockerLabs\backup\:/backup ubuntu tar xzf /backup/student-data.tar.gz -C /data

D:\>docker run --rm -v restored-data:/data ubuntu ls -l /data
total 12
-rw-r--r-- 1 root root 23 Aug  8 05:37 container1.txt
-rw-r--r-- 1 root root 24 Aug  8 05:35 notes.txt
-rw-r--r-- 1 root root 23 Aug  8 05:36 student.txt

D:\>|
```

```
C:\Windows\System32\cmd.exe x + v
195bb2fc9bce attendance-management-backend "docker-entrypoint.s-" 4 days ago Up About an hour (healthy) 0.0.0.0:5000->5000/tcp, [::]:5000->
5000/tcp attendance_backend
b1841cfc7cb mongo:7 "docker-entrypoint.s-" 4 days ago Up About an hour (healthy) 0.0.0.0:27017->27017/tcp, [::]:2701
7->27017/tcp attendance_mongo

D:\>

D:\>mkdir C:\DockerLabs\backup

D:\>docker run --rm -v student-data:/data -v C:\DockerLabs\backup\:/backup ubuntu tar czf /backup/student-data.tar.gz -C /data .

D:\>docker volume create restored-data
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D:\>docker run --rm -v restored-data:/data -v C:\DockerLabs\backup\:/backup ubuntu tar xzf /backup/student-data.tar.gz -C /data

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total 12
-rw-r--r-- 1 root root 23 Aug 8 05:37 container1.txt
-rw-r--r-- 1 root root 24 Aug 8 05:35 notes.txt
-rw-r--r-- 1 root root 23 Aug 8 05:36 student.txt

D:\>docker volume create student-records
student-records

D:\>docker run -dit --name student-app -v student-records:/data ubuntu
47c96030e633c29417eac7cbd26613a8abea492b38c39d4b02dbece96ded6f47

D:\>docker exec student-app bash -c "echo '101, Rahul, BCA' > /data/students.csv"

D:\>docker exec student-app bash -c "echo '102, Priya, BCA' >> /data/students.csv"

D:\>docker exec student-app cat /data/students.csv
101, Rahul, BCA
102, Priya, BCA

D:\>|
```

```
C:\Windows\System32\cmd.exe x + v

D:\>docker run --rm -v student-data:/data -v C:\DockerLabs\backup\:/backup ubuntu tar czf /backup/student-data.tar.gz -C /data .

D:\>docker volume create restored-data
restored-data

D:\>docker run --rm -v restored-data:/data -v C:\DockerLabs\backup\:/backup ubuntu tar xzf /backup/student-data.tar.gz -C /data

D:\>docker run --rm -v restored-data:/data ubuntu ls -l /data
total 12
-rw-r--r-- 1 root root 23 Aug 8 05:37 container1.txt
-rw-r--r-- 1 root root 24 Aug 8 05:35 notes.txt
-rw-r--r-- 1 root root 23 Aug 8 05:36 student.txt

D:\>docker volume create student-records
student-records

D:\>docker run -dit --name student-app -v student-records:/data ubuntu
47c96030e633c29417eac7cbd26613a8abea492b38c39d4b02dbece96ded6f47

D:\>docker exec student-app bash -c "echo '101, Rahul, BCA' > /data/students.csv"

D:\>docker exec student-app bash -c "echo '102, Priya, BCA' >> /data/students.csv"

D:\>docker exec student-app cat /data/students.csv
101, Rahul, BCA
102, Priya, BCA

D:\>docker rm -f student-app
student-app

D:\>docker run -dit --name student-app-new -v student-records:/data ubuntu
a74d8debc8dd5b18599e46dcf936507eb7e979f8d803f85ff7bee785f3a739

D:\>docker exec student-app-new cat /data/students.csv
101, Rahul, BCA
102, Priya, BCA

D:\>|
```